SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 **Total Marks:50**

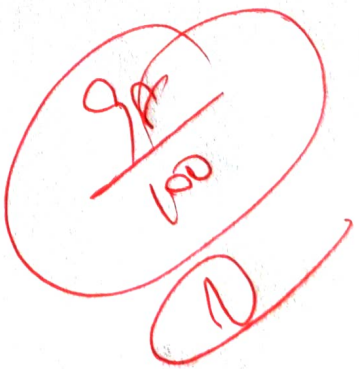
Annexure A

CONCEPT MAPPING

Code of Conduct:

I Sanjana / 192126007 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Sanjana.
Signature of the student:



Annexure A

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:
    a. Show how each attack leads to specific threats and risks
    b. Map how security goals help mitigate them Provide one real-world example for each link

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
    a. Show differences and similarities
        i.Map how each affects the Data structure and Security level
        ii.Analyze which is more vulnerable to attacks and why

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
    a. Show relationships between these concepts
    b. Analyze how each contributes to secure communication
    c. Include one real-world application for each

5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
    a. Show how each concept is mathematically connected
    b. Map their role in encryption/decryption
    c. Analyze which concept is most critical for public key cryptography and justify

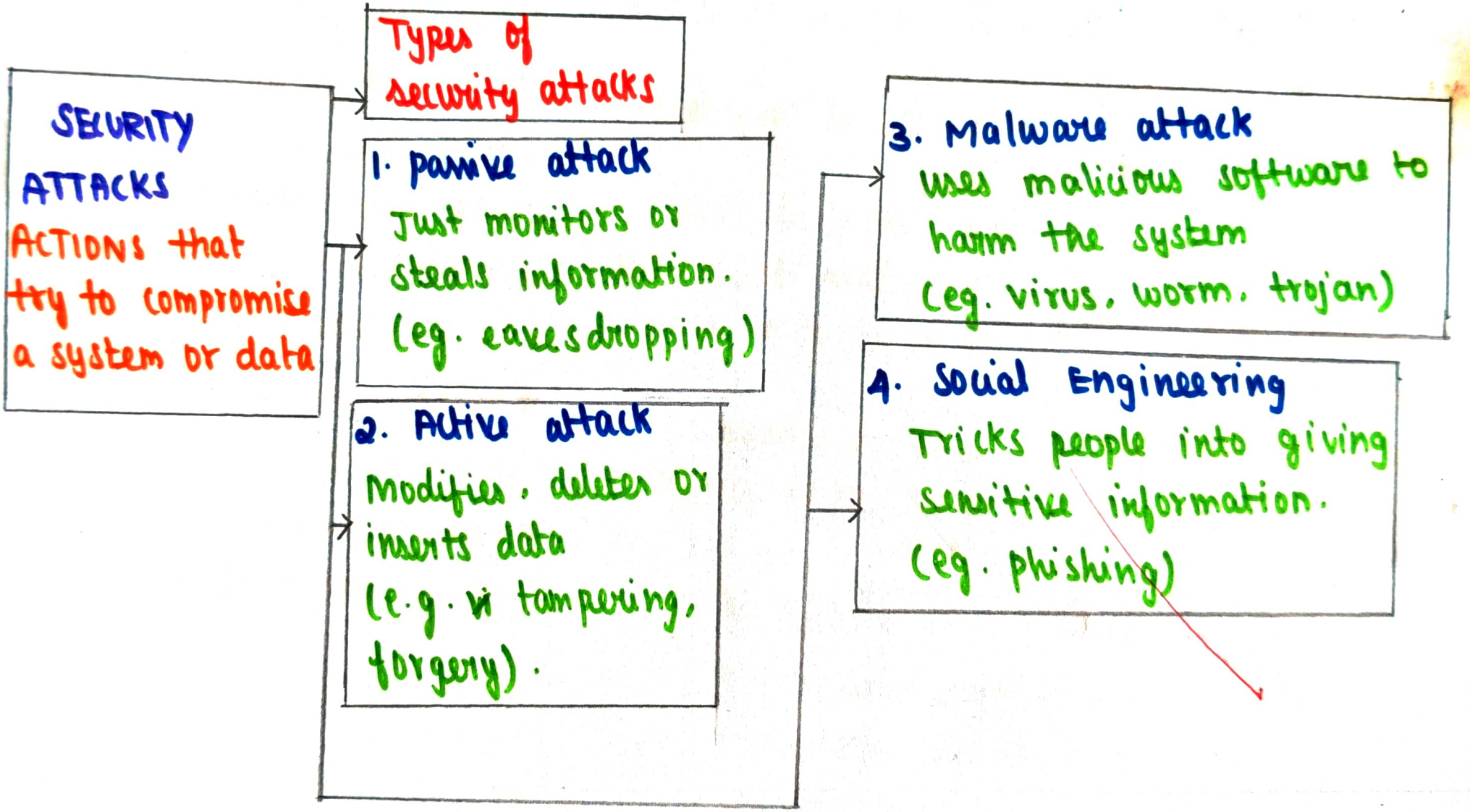
RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

27
30
20
15
10
88
18/100
2

CONCEPT MAPPING

1) SECURITY ATTACKS, THREATS, RISKS AND SECURITY Goals.



Threats

potential dangers that can exploit vulnerabilities

* Unauthorized access
* Data leakage
* Malwar infection
* Insider threats
* phishing and fraud
* physical threats (theft, loss, damage)

RISKS

The possible impact if a threat is success-ful.

* Loss of confidential data
* financial loss
* system downtime
* Reputation dama-ge.
* Legal issues
* Loss of customer user.

Security Goals

What we aim to protect.

* Confidentiality (Keep data private)
* Integrity (Keep data accurate)
* Availability (Keep system accessible)
* Authentication (verify identity)
* Non-repudiation (prevent denial of action)

b) How Security Goals Help mitigate them with real world examples

Security Goals What we aim to protect	How it helps mitigate risks (and against which threats)
Confidentiality (Keep data private)	* Encryption of data * Access control and authentication * keep sensitive data hidden from attackers.
Integrity (Keep data accurate)	* Hashing and digital signatures * Input validation * prevents data from being altered.
Availability (Keep systems accessible)	* Backup and recovery systems * firewalls and intrusion detection * Reduces downtime and ensure continuous access.

Real world Examples

(How security goals reduce the impact)

* Online banking - Encryption protects account details
* Healthcare - keeps patient records private

* E-commerce - digital signatures ensure payment data is not changed.
* Social media - prevents tampering with posts

* Cloud services - backup keeps data safe during server failure
* Company record - firewalls prevent Dos attack and keep services running.

2) # classical cipher Techniques

(Caesar cipher → monoalphabetic cipher → vigener cipher → Hill cipher)

Caesar cipher

A simple substitution cipher where each letter is shifted by a fixed number of positions in the alphabet.

Example : (shift = 3) :

HELLO - KHOOR (each letter +3)

Key usage : single integer

Complexity : very low

Security strength : weak (easy to break).

Monoalphabetic cipher

Each letter of the plantext is replaced by a fixed substitute letter from another alphabet.

Example : (simple substitution) :

A → Q B → W C → E ... (fixed mapping for all letters).

Key usage : substitution alphabet

Complexity : Low to medium

Security strength : moderate (can be broken using frequency analysis).

Vigenere cipher

uses a repeating Keyword to shift letters. The shift value changes for each letter based on the key word.

Example (Key : KEY)

ATTACK → KFDLGO (Shift : 10, 4, 23, 0, 2, 6)

Key usage : Keyword

complexity : Medium

Security strength : stronger than monoalphabetic

Hill cipher

uses matrix multiplication with a key matrix to encrypt blocks of letters (more than one letter at a time).

Example (2×2 matrix)

plaintext HI (8, 9)

Key : $\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$ ciphertext : VM (21, 12)

Key usage : matrix with numbers

complexity : high

Security strength : very strong (more difficult to break, relies on linear algebra).

Comparison at a glance

Feature	Ceasar cipher	Monalphabetic	Vigenere	Hill cipher
Key usage	Single integer	Substitution letters	Keyword	Matrix with numbers
Complexity	Very low	Low to medium	Medium	High
Security strength	Weak	Moderate	Stronger than monoalphabetic	Very strong

Hill cipher → (due to matrix-based encryption and higher complexity).